

Feature Perturbation vs. Selection Shift: Post-hoc Calibration of Classical Tabular Models

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Abstract

Temperature scaling and related post-hoc maps are often treated as a cheap fix for miscalibrated probabilities. We measure whether that fix transfers under two different test-time changes, for logistic regression, random forests, and histogram gradient boosting on small tabular datasets (no deep nets, no API models). *Feature mean-shift with frozen labels*—a perturbation of X that does **not** preserve $P(Y | X)$ —raises ECE: at strength 1.5, $\Delta\text{ECE} = \text{ECE}_{\text{pert}} - \text{ECE}_{\text{iid}}$ is positive in 51/60 seed-level cells without calibration (mean 0.113, cell-to-cell sd 0.149) and remains positive after i.i.d. temperature, isotonic, or histogram maps. Averaging seeds first, all 12/12 dataset $\times$ model means are positive (exploratory Wilcoxon $p=2.4\text{e-}04$). The same maps can help or hurt relative to doing nothing. By contrast, *selection* on X that keeps the original (X, y) pairs—closer to covariate shift—yields much smaller ΔECE (quantile-slice cell-mean 0.016, sd 0.006). Unweighted split-conformal coverage falls under feature perturbation. This is an analysis result, not a new calibrator.

1 Introduction

A classifier is *calibrated* when a predicted probability p is correct with frequency p . Post-hoc maps—Platt-style scaling, histogram binning, isotonic regression, and temperature scaling—are the standard way to improve calibration without retraining [1–3]. A separate literature shows that deep-net uncertainty *collapses* under dataset shift [4–6]. Almost all of that evidence is vision-scale neural nets. Tree ensembles remain strong on tabular data [7], and their i.i.d. calibration properties have been known for decades [2]. Within our API-verified set of 47 papers—mostly deep-net calibration and conformal theory, not a systematic search of applied tabular-drift venues—we did not find this exact measurement (sklearn models, i.i.d. post-hoc maps, test-time feature perturbation vs. selection). That is a scoped gap, not a proof of absence.

We treat that as a measurement question. We do not introduce a new calibrator. A negative or mixed answer is an acceptable outcome.

Primary hypothesis (reframed). I.i.d.-fitted post-hoc maps do not keep ECE from rising under a Gaussian mean-shift of test features with *labels held fixed* (input perturbation / a change in $P(Y | X)$ at the observed x). We do **not** claim that genuine covariate shift—selection on X that preserves pairings—has the same effect; that is tested as a control.

Two other hypotheses were piloted and *not* selected as the paper claim: isotonic overfit vs. temperature as a function of n_{cal} (H2; heterogeneous), and oracle-weighted conformal recovery (H3; the 1-D weight estimator failed). Decision log: logs/decisions.md.

2 Related work

Surveys of calibration methods and metrics include [8, 9]. ECE via confidence binning [3, 10] is biased and definition-dependent [11–13]; we therefore also report Brier score [14] and NLL [15]. Temperature scaling [3] and Dirichlet calibration [16] target neural nets. Unsupervised recalibration under covariate shift exists for neural models [17]. Domain-drift post-hoc calibration of CNNs is studied by [6]. Those papers already *predict* our headline outcomes: ECE should rise under shift, and i.i.d.-fitted post-hoc maps should not be a complete fix. A surprising result here would have been that logistic regression, random forests, and histogram boosting *stay* calibrated under a mean-shift of used covariates *with labels frozen*. Selection that preserves (X, y) pairs is a different mechanism. We reuse the questions, not the model class.

Conformal prediction has finite-sample coverage under exchangeability [18–20] and requires density-ratio weights under covariate shift [21]. We use unweighted split conformal as a secondary diagnostic, not as a proposed fix.

3 Methods

Models. Scikit-learn logistic regression (with standard scaling), random forests, and HistGradientBoostingClassifier, with hyperparameters as in src/calibshift/models.py. All expose predict_proba.

Calibrators. Fitted on a held-out calibration split of the *source* distribution, then frozen. `none`: identity. `temperature`: one scalar T on $\log p$, fitted by NLL (Guo-style, using log-probabilities so trees without logits still apply). `isotonic`: sklearn isotonic regression on $p(y=1)$ (binary) or on confidence (multiclass). `histogram`: equal-width binning of the same 1-D score.

Shifts. *Feature perturbation*: add $s \cdot \hat{\sigma}_j$ to listed test columns, **keeping the original labels**. This moves x without moving y , so it is *not* covariate shift ($P(Y | X)$ is not invariant). $\hat{\sigma}$ is the test-split feature sd. *Selection*: quantile slice and importance resampling of existing (X, y) pairs (closer to covariate shift). Trailing-column perturbation is a control.

Metrics. Equal-width confidence ECE with 15 bins (from the exp04 config), Brier, NLL, accuracy, and split conformal coverage with the inverse-probability (LAC) score $1 - p_y$ at $\alpha = 0.1$ (not APS).

Protocol. Train/calibration/test partitions follow `calibshift.runner` (source-distribution splits; calibration data never includes the test set), stratified, with the 5 seeds listed in the exp04 config. Results are written only by `calibshift.io.write_result`. `paper/numbers.tex` is generated from those JSON files.

4 Experiments

Datasets. Offline-only: sklearn `breast_cancer` (binary) and `wine` (3-class); two `make_classification` sets (binary and 3-class). No OpenML, no paid APIs.

Main grid (exp04_main_h1). 240 cells, 0 failed, 23.0 seconds on CPU. This is the source of all headline numbers.

Ablations. exp05: quantile slice and importance resampling. exp06: calibration-set size (killed primary H2). exp07: shift trailing columns.

Sanity. I.i.d. logistic regression on breast-cancer: accuracy 0.979, ECE 0.032 ± 0.010 (linear models on this set are typically $\gtrsim 0.95$ accurate). Wine random forests are overconfident i.i.d. (ECE 0.147 ± 0.020), matching the classical tree-calibration picture [2]. Split conformal coverage at $s=0$ is near the nominal $1 - \alpha = 0.9$ for $\alpha=0.1$. We do *not* compare ECE magnitudes to ImageNet temperature scaling [3].

5 Results

5.1 Feature perturbation usually raises ECE

Figure 1 plots shifted ECE against perturbation strength s . Figure 2 shows mean ΔECE at $s=1.5$. Pooled over 60 dataset $\times$ model $\times$ seed cells, uncalibrated mean $\Delta\text{ECE} =$

Table 1: Seed-pooled ΔECE at feature-perturbation $s=1.5$ ($n = 60$ cells). “std” is heterogeneity, not se of the mean. Wilcoxon p on the 60 cells is exploratory; a seed-averaged $n = 12$ test is $p=2.4\text{e-}04$. 21 tests in `results/stats_tests.json`, unadjusted.

Calibrator	mean ΔECE	std	#pos/neg	Wilcoxon p
none	0.113	0.149	51/9	2.2e-10
temperature	0.124	0.150	56/4	3.1e-11
isotonic	0.141	0.151	53/7	4.6e-11
histogram	0.141	0.151	57/3	3.3e-11

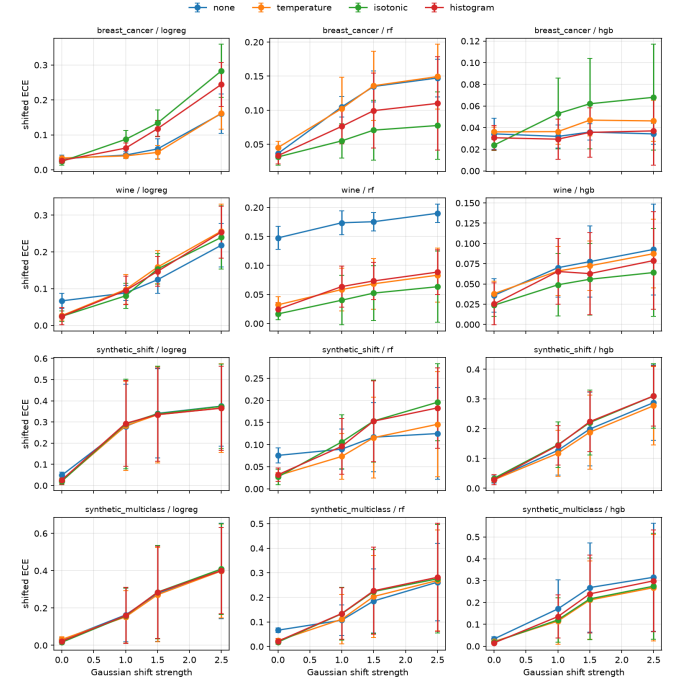


Figure 1: Shifted ECE vs. Gaussian feature-perturbation strength (mean $\pm$ seed sd). $s=0$ is i.i.d. Labels are frozen; this is not covariate shift. Source: `results/exp04_main_h1.json`.

0.113 with cell-to-cell sd 0.149 (51 positive, 9 negative). After i.i.d. temperature scaling the pooled mean is 0.124 (sd 0.150). These sd figures describe heterogeneity, not a confidence interval for the mean. Averaging seeds first, all 12 of 12 dataset $\times$ model means are positive (exploratory Wilcoxon $p=2.4\text{e-}04$; still not independent across models that share a dataset). Breast-cancer histogram boosting is the smallest cell (mean 0.002, seed sd 0.009). Synthetic logistic regression is the largest (mean 0.292, sd 0.214) and coincides with a large accuracy drop. Table 1 records seed-pooled statistics; `results/stats_tests.json` contains 21 related p -values with no multiplicity correction.

5.2 Post-hoc maps are not a reliable remedy

Figure 3 plots $\text{ECE}_{\text{shift(cal)}} - \text{ECE}_{\text{shift(none)}}$. Wine random forests *benefit* from isotonic (-0.123 ECE; shifted ECE

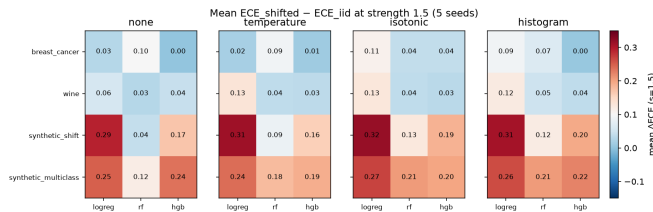


Figure 2: Mean ΔECE at $s=1.5$. Positive (red) means the shifted test is *worse* calibrated than the i.i.d. test.

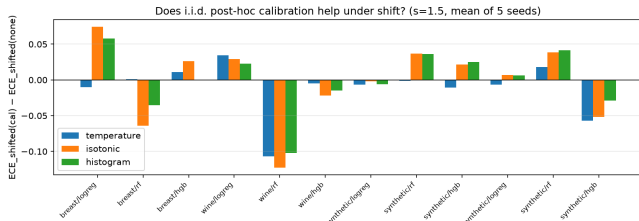


Figure 3: Does i.i.d. calibration help on the shifted test? Negative bars mean the calibrator beats *none* under shift.

0.175 $\rightarrow$ temperature 0.068), because the uncalibrated forest was already badly calibrated. Breast-cancer logistic regression is *hurt* by isotonic (0.074). We therefore do not rank calibrators under shift.

5.3 Coverage under perturbation

Unweighted split conformal (logreg, *none*, LAC score $1 - p_y$) coverage at $\alpha = 0.1$: breast-cancer 0.913 ($s=0$), 0.778 ($s=1.5$), 0.608 ($s=2.5$); synthetic binary 0.634 at $s=1.5$ (Figure 4). Exchangeability is broken by construction. A 1-D “oracle” density ratio on feature 0 did *not* restore coverage in the H3 pilot (killed).

5.4 Selection shift is a different, weaker effect

Quantile slicing and importance resampling keep original (X, y) pairs, so $P(Y | X)$ is preserved and the mechanism is closer to covariate shift. Mean ΔECE across dataset $\times$ model cells (*none*) is 0.016 (sd 0.006) for quantile slice and 0.011 (sd 0.031) for importance resampling—an order of magnitude below the feature-perturbation pooled mean 0.113. Importance resampling can even *decrease* ECE on breast-cancer. This control **does not support** a claim that i.i.d. post-hoc calibration fails under covariate shift in general. Trailing-column perturbation on *synthetic_shift* yields RF $\Delta ECE = 0.009$ vs. 0.041 when perturbing the first four columns.

6 Limitations

- **The headline mechanism is not covariate shift.** Frozen-label feature mean-shift changes $P(Y | X)$ at the observed x (input corruption). Selection controls that pre-

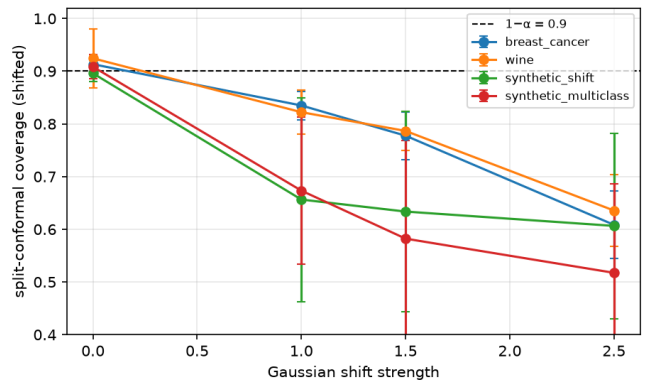


Figure 4: Unweighted split-conformal coverage vs. feature perturbation (logreg, no calibrator, LAC score). Dashed line: $1 - \alpha = 0.9$.

serve pairings show small ΔECE . Title and abstract are scoped accordingly.

- **Shift is synthetic** and, for feature perturbation, entangled with accuracy drop (synthetic logreg 0.932 $\rightarrow$ 0.620 at $s=1.5$).
- **Small tabular data only.** Wine test splits are ~ 45 points with 15 ECE bins; those cells are noisy. No ImageNet, no LMs, no naturally occurring drift (no Folktables/WILDS).
- **Multiclass isotonic/histogram** map confidence, not a full Dirichlet map [16]; argmax can flip after renormalization.
- **Temperature on probabilities.** Trees have no logits; RF 0/1 probabilities saturate after clipping. This is not Guo logit temperature.
- **H3 weights were misspecified.** A 1-D histogram ratio is not $p_{\text{test}}(x)/p_{\text{cal}}(x)$ for a 4-D perturbation.
- **Pooling and tests.** Seed-level $n = 60$ Wilcoxon is pseudo-replicated. We also report $n = 12$ seed-averaged dataset $\times$ model means. 21 tests, unadjusted.
- **Search scope.** Novelty is relative to paper/verified.bib (47 API-verified papers).

7 Conclusion

For these sklearn models, i.i.d. post-hoc calibration is not a dependable answer to *test-time feature perturbation with frozen labels*: ECE usually rises and conformal coverage falls, while the maps can make perturbed ECE better or worse than doing nothing. The same experiments **do not** show a comparable failure under selection on X . The HGB exception and the killed H2/H3 hypotheses are part of the result.

AI-generation disclosure

This repository and draft were produced by a coding agent executing the user's GOAL.md protocol (literature verification via arXiv and Semantic Scholar APIs, experiments on CPU, no hand-edited results/*.json). Every numeric macro in paper/numbers.tex is generated from those files. Citations are restricted to paper/verified.bib. A human should still read the limitations and re-run make fresh-clone-test before treating the PDF as camera-ready.

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